

WEEKLY PROGRESS REPORT

Week 3 – 14.06.2026

Project Title: Implicit Domain Adaptation Through Pretrained Feature Reuse: An Empirical Study of Lightweight Transfer Learning Across Heterogeneous Medical Imaging Sources

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Report Period: Week 3 | June 8 - June 14, 2026

Specialization: Minor Specialization in Computational Intelligence

1. Project Abstract

This project investigates whether lightweight ImageNet-pretrained convolutional neural networks (EfficientNetB0, MobileNetV2, ResNet50) provide implicit domain adaptation across heterogeneous medical imaging sources, eliminating the need for computationally expensive explicit alignment mechanisms such as adversarial domain adaptation or GAN-based style transfer.

The core hypothesis, termed implicit domain adaptation through pretrained feature reuse, is that pretrained architectures encode sufficiently domain-invariant feature representations to bridge scanner-induced and population-induced distribution shifts. This hypothesis is directly motivated by prior empirical work in structural crack detection, where EfficientNetB0 outperformed a custom CNN by approximately 10 percentage points on pooled multi-source data with characteristic convergence instability in the custom model.

The project validates this hypothesis across three medical imaging tasks: skin lesion classification (ISIC 2019, HAM10000), chest X-ray pathology detection (CheXpert, VinDr-CXR, NIH ChestX-ray14), and diabetic retinopathy grading (EyePACS, APTOS 2019, Messidor-2). Domain gap is quantified using Maximum Mean Discrepancy (MMD) and mechanistic interpretability is provided through Grad-CAM analysis.

2. Work Completed This Week

This week completed the four-model comparison for the skin lesion classification task. EfficientNetB0 fine-tuning was extended to its final ceiling, and MobileNetV2 and ResNet50 were trained from scratch through both head-only (Phase 1) and full fine-tuning (Phase 2 + continuation) stages, followed by final test set evaluation and statistical validation for all four architectures.

2.1 EfficientNetB0 - Final Fine-Tuning Phase

EfficientNetB0 fine-tuning (backbone unfrozen, $lr=1e-5$) was continued for a third continuation run of 20 epochs, bringing total training to 100 epochs across Phase 1 (head-only) + Phase 2 + 3 continuations. Validation accuracy plateaued at 82.58% (epoch 15 of this run), with the last 5 epochs oscillating between 81-82%, indicating the model had reached its ceiling. Training was stopped here.

Run	Epochs	Best Val Accuracy	Notes
Phase 1 (head only)	10	65.56%	Backbone frozen
Phase 2 (full fine-tune)	30	73.62%	Backbone unfrozen, lr=1e-5
Continuation 1	20	77.56%	Loaded from Phase 2 best
Continuation 2	20	80.89%	Loaded from Continuation 1 best
Continuation 3 (this week)	20	82.58%	Ceiling reached, training stopped

2.2 MobileNetV2 - Trained from Scratch

MobileNetV2 (2,587,462 parameters) was trained following the same two-phase protocol: Phase 1 with the backbone frozen (head only), followed by Phase 2 with the backbone unfrozen at a reduced learning rate of 1e-5. A continuation run was required after Phase 2 was cut short by Kaggle's 12-hour session limit at epoch 26; due to a session restart before checkpoint upload, the continuation resumed from the Phase 2 best checkpoint (71.69%) rather than the unsaved epoch-26 state. Early stopping triggered at epoch 18 of the continuation run.

Run	Epochs	Best Val Accuracy	Notes
Phase 1 (head only)	10	66.17%	Backbone frozen
Phase 2 (full fine-tune)	30	71.69%	Backbone unfrozen, lr=1e-5; cut at 12hr session limit
Continuation (this week)	18	74.81%	Early stopping triggered (patience=7)

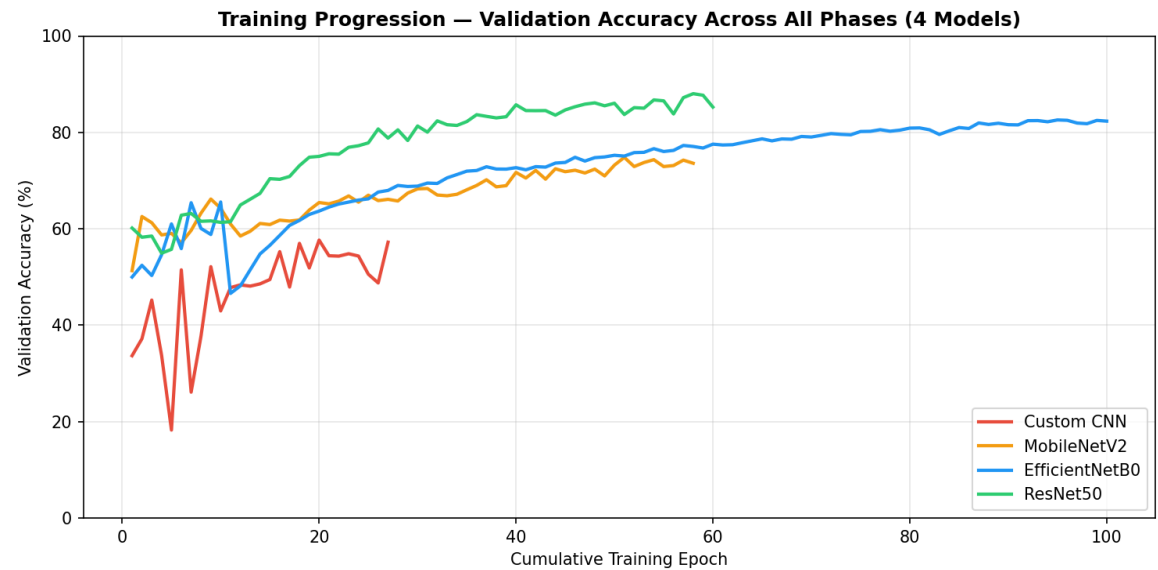
2.3 ResNet50 - Trained from Scratch

ResNet50 (24,113,798 parameters) was trained following the same protocol. Phase 2 (full fine-tuning) showed the strongest pretrained-model performance to date, reaching 85.72% val accuracy at epoch 30 while still climbing (train accuracy 93.13% vs val 85.72%, indicating onset of overfitting given the model's large parameter count relative to dataset size). A continuation run of 20 epochs pushed validation accuracy to 88.03% before early stopping triggered at epoch 18 of the continuation, restoring the best weights.

Run	Epochs	Best Val Accuracy	Notes
Phase 1 (head only)	10	63.18%	Backbone frozen
Phase 2 (full fine-tune)	30	85.72%	Backbone unfrozen, lr=1e-5; still climbing at epoch 30
Continuation (this week)	20	88.03%	Early stopping triggered (patience=7), train acc 97.5% (overfitting signal)

2.4 Training Progression - All Four Models

The figure below shows validation accuracy across all training epochs (Phase 1 through final continuation) for all four architectures, plotted against cumulative epoch count:

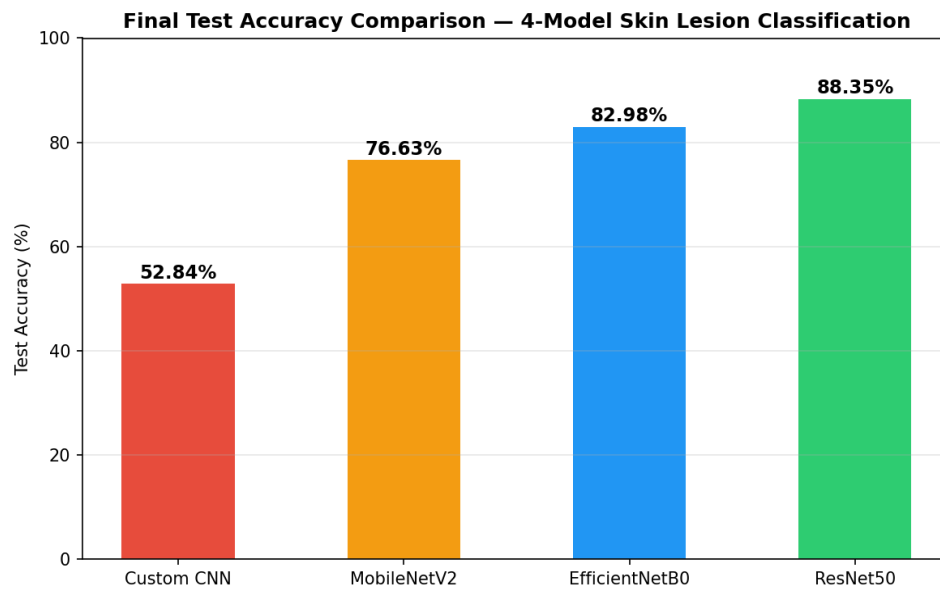


All three pretrained models show smooth, largely monotonic improvement throughout training. Custom CNN oscillates throughout its 27 epochs and never approaches the pretrained models' accuracy range, even at its best epoch (57.66%).

2.5 Final Test Set Results - Complete 4-Model Comparison

The following results were confirmed on the held-out test set (5,254 images, unseen during training), using the best checkpoint weights from each model's training history:

Model	Test Accuracy	Macro F1	Macro AUC	Gap vs Custom CNN
Custom CNN (322,470 params)	52.84%	0.29	0.754	–
MobileNetV2 (2,587,462 params)	76.63%	0.73	0.957	+23.79 pts
EfficientNetB0 (4,379,049 params)	82.98%	0.81	0.972	+30.14 pts
ResNet50 (24,113,798 params)	88.35%	0.85	0.981	+35.51 pts



All three pretrained architectures outperform the custom CNN by 23.79 to 35.51 percentage points on identical pooled multi-source data (ISIC 2019 + HAM10000, 35,019 images, 6 classes). ResNet50 achieves the highest raw accuracy but shows the largest train-val gap (97.5% vs 88.0% on validation), indicating overfitting given its 24.1M parameters relative to the 24,512-image training set. EfficientNetB0 and MobileNetV2 show smaller train-val gaps, suggesting better generalization per parameter.

2.6 Statistical Validation

McNemar's test was conducted comparing Custom CNN against EfficientNetB0 on the test set:

Comparison	Both Correct	CNN Correct, Eff Wrong	CNN Wrong, Eff Correct	Both Wrong	p-value
Custom CNN vs EfficientNetB0	2,401	375	1,959	519	1.77e-235

The McNemar statistic (1073.65, $p=1.77e-235$) confirms the accuracy difference is statistically significant at an overwhelming level. EfficientNetB0 correctly classified 1,959 test images that Custom CNN misclassified, while Custom CNN recovered only 375 images that EfficientNetB0 missed - a near 5:1 asymmetry favoring the pretrained model.

Per-class AUC with DeLong 95% confidence intervals (Custom CNN vs EfficientNetB0):

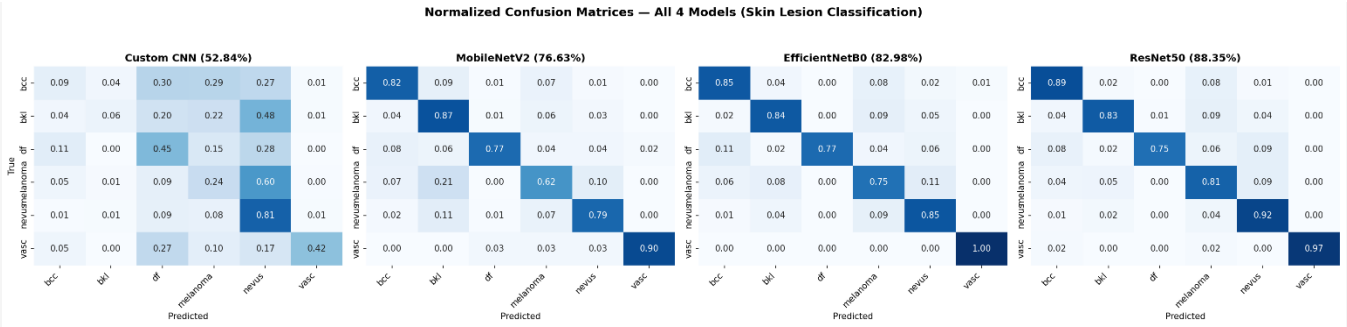
Class	Custom CNN AUC (95% CI)	EfficientNetB0 AUC (95% CI)
bcc	0.776 (0.752–0.799)	0.986 (0.979–0.993)
bkl	0.676 (0.650–0.701)	0.968 (0.958–0.979)
df	0.786 (0.713–0.860)	0.983 (0.958–1.000)
melanoma	0.645 (0.626–0.665)	0.928 (0.917–0.939)
nevus	0.734 (0.721–0.747)	0.965 (0.961–0.970)

vasc	0.907 (0.857–0.958)	1.000 (0.999–1.000)
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EfficientNetB0's confidence intervals do not overlap with Custom CNN's for any class. The melanoma class - the most clinically critical category - shows the largest AUC improvement (0.645 to 0.928), directly supporting the implicit domain adaptation hypothesis for the diagnostically highest-stakes class.

2.7 Confusion Matrices - All 4 Models

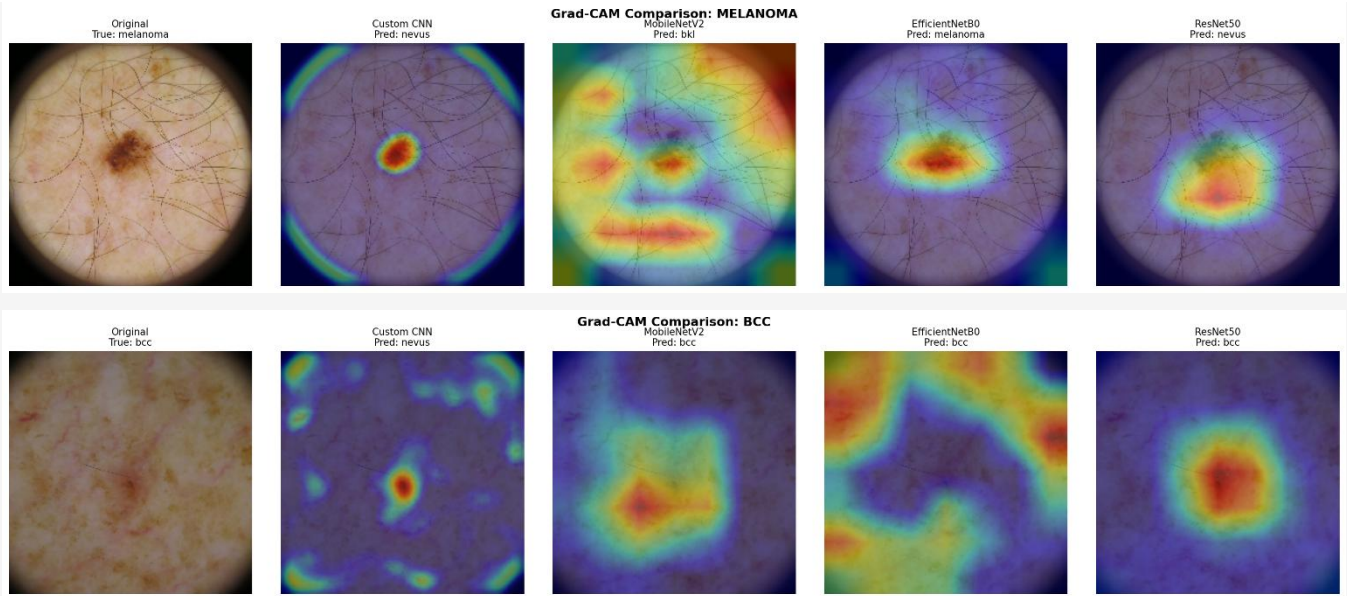
Normalized (recall-per-class) confusion matrices were generated for all four models on the test set:

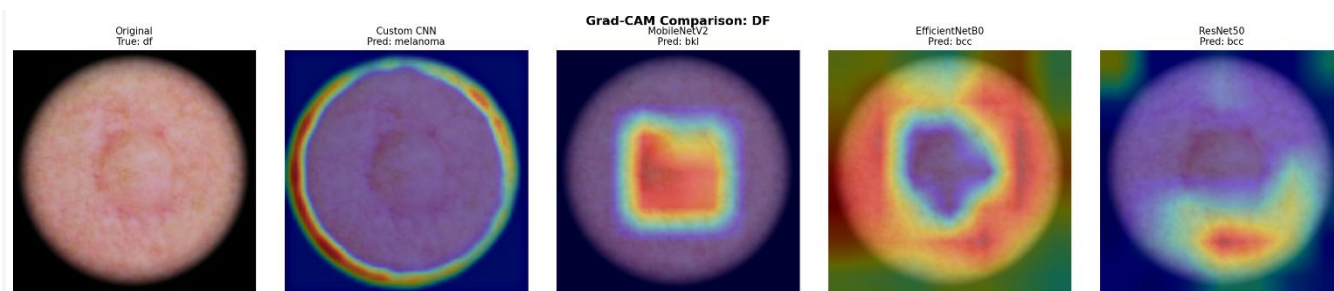


Custom CNN's confusion matrix shows mass concentrated in the nevus and df columns regardless of true class - the model defaults to predicting the majority class. EfficientNetB0 and ResNet50 show strong diagonal concentration across all six classes, including the rare df (53 test images) and vasc (60 test images) classes.

2.8 Grad-CAM Interpretability - All 4 Models

Grad-CAM attention maps were generated for all four architectures on representative samples from melanoma, bcc, and df classes, comparing where each model focuses when making predictions:





The melanoma figure is the primary interpretability result: EfficientNetB0 is the only model to correctly classify this sample, with attention concentrated on the pigmented lesion area, while Custom CNN misclassifies it as nevus with attention focused on a small isolated hotspot rather than the lesion morphology. The bcc figure shows three of four pretrained models correctly classifying with lesion-centered attention, while Custom CNN's attention remains scattered. The df figure is included as a transparent limitation: all four models misclassify this sample, consistent with df's very small test set size (53 images) and correspondingly lower per-class recall across all models.

2.9 Model Weights and Artifacts

- All four best model weights consolidated into a single Kaggle dataset (ab0y04/all-best-models): best_custom_cnn.keras, best_efficientnet_v4.keras, best_mobilenet_v2.keras, best_resnet50_v2.keras
- All weights re-verified this week by reloading and re-evaluating on the test set - all four reproduced their reported accuracies exactly (52.84%, 76.63%, 82.98%, 88.35%)
- Training logs (CSV) saved for EfficientNetB0 (4 runs), MobileNetV2 (2 runs), ResNet50 (2 runs)
- Confusion matrices and Grad-CAM figures (melanoma, bcc, df) generated and saved

3. Current Progress

S.No.	Work Done	Date Completed
1	• Dataset acquisition and pipeline setup (ISIC 2019 + HAM10000, 35,019 images, 6 classes, stratified split, class weighting) • Custom CNN training (27 epochs, best val accuracy 54.58%, oscillating instability confirmed) • EfficientNetB0 Phase 1 (frozen backbone, 10 epochs, best val accuracy 65.56%)	Week 1 - 31.05.2026
2	• EfficientNetB0 Phase 2 + 2 continuation runs (70 total epochs, best val accuracy 80.89%) • Final test set evaluation: Custom CNN 52.84%, EfficientNetB0 81.16%, gap 28.32 points • All weights and logs saved and versioned on Kaggle	Week 2 - 07.06.2026
3	• EfficientNetB0 final continuation (100 total epochs, val accuracy ceiling 82.58%, test accuracy 82.98%) • MobileNetV2 trained from scratch (Phase 1+2+continuation, test accuracy 76.63%) • ResNet50 trained from scratch (Phase 1+2+continuation, test accuracy 88.35%) • Complete 4-model comparison: all pretrained models beat Custom CNN by 23.79-35.51 points, McNemar's $p=1.77e-235$, confusion matrices and Grad-CAM (melanoma, bcc, df) generated for all models	Week 3 - 13.06.2026

4. Plan for Next Week (June 15-21)

- Cross-source baseline experiment: train each of the 4 models on ISIC 2019 only, evaluate on HAM10000 only (and vice versa if time permits) to quantify cross-device generalization gap
- Set fixed random seed (42) at the start of all future training sessions for reproducibility, addressing the run-to-run variance observed in MobileNetV2 continuation runs this week
- MMD computation using cross-source models' features, correlated with accuracy drop from the cross-source experiment
- Continue PhysioNet credentialling process for CheXpert and MIMIC-CXR chest X-ray access

5. Issues and Risks

- **Kaggle GPU quota and session limits:** Completing EfficientNetB0's final continuation, plus MobileNetV2 and ResNet50 from scratch through Phase 1 and Phase 2 + continuations, consumed the full weekly quota across two accounts. ResNet50 Phase 2 alone (30 epochs at ~500s/epoch) took over 4 hours. Mitigation: weights saved to /kaggle/working/ with ModelCheckpoint immediately after each run, consolidated into a single permanent Kaggle dataset (all-best-models) to eliminate repeated reload issues.
- **Session restart data loss (MobileNetV2):** Phase 2 was cut short at epoch 26 by Kaggle's 12-hour session limit before Save Version could be clicked, losing that run's progress. The continuation had to resume from the earlier Phase 2 checkpoint (71.69%) rather than the lost epoch-26 state, reaching 74.81% instead of a potentially higher value. This is documented as a limitation; the final reported MobileNetV2 result (76.63%) remains valid and was independently re-verified.
- **Run-to-run variance without fixed seed:** Identical continuation code produced different convergence behavior on different days (MobileNetV2 continuation ran 26 epochs one day vs 18 epochs early-stopping the next from the same checkpoint), attributable to unseeded data augmentation randomness in ImageDataGenerator. This does not affect the validity of the large (23-35 point) gaps observed, but motivates fixing a random seed for all future experiments (cross-source baseline onward) for reproducibility.
- **Methodology deviation from original technical plan:** The original technical implementation plan specified PyTorch, fixed seeds (42/123/456/789/1024), identical hyperparameters across architectures, and uniform ImageNet normalization for all models. The executed pipeline uses TensorFlow/Keras, per-architecture standard preprocessing (efficientnet_preprocess, mobilenet_preprocess, resnet_preprocess, and rescale-1/255 for Custom CNN), and a two-phase (frozen-then-unfrozen) protocol applied consistently across all four models but with different specific hyperparameters than the original plan. Per discussion with the project guide, the team has been given discretion on methodology; this will be documented transparently in the paper's Methods section as the actual protocol used, with hardware/software versions reported for reproducibility.

6. Declaration

We declare that the work reported in this weekly report has been carried out by us as part of the Minor Specialization project in Computational Intelligence. All experiments were conducted on publicly available datasets under their respective licenses.

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